An Audited Evaluation of Explainable RF–XGBoost Stacking for Multiclass Sleep-Disorder Classification on a Small Public Benchmark

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Abstract

Small public tabular health datasets are easily over-interpreted. We audit the Kaggle Sleep Health and Lifestyle Dataset (374 records, three outcome classes) and re-evaluate Random Forest–XGBoost (RF–XGB) stacking with a logistic meta-learner under leakage-controlled, repeated validation. Opening the source file shows that the 219 "no disorder" records carry the literal label "None"; treating them as missing, as in our earlier draft, was an import artefact. The file contains 242 exact duplicate rows after the identifier is removed (132 unique rows, 109 distinct predictor vectors); 22 predictor vectors carry conflicting labels, which caps attainable accuracy on the full file at 93.3%. Under duplicate-aware 10×5-fold cross-validation the stack reached 89.6% accuracy (95% cluster-bootstrap CI 86.1–92.7) and macro-F1 0.87, statistically indistinguishable from soft voting (89.8%), random forest (89.8%), XGBoost (89.8%) and plain logistic regression (89.7%); the majority-class baseline scored 58.6%. Plain stratified splitting that separates duplicate rows inflated stack accuracy by 2.2 points, and on the 132 unique rows alone accuracy fell to 73.0%. A re-implementation of the original single 75-row hold-out gave 72/75 correct (the previously reported 73/75 could not be reproduced), and 68 of those 75 test rows had an identical predictor vector in the training set. SHAP rankings were stable across background resamples within an architecture (rank correlation 0.98 for stacking, 0.97 for soft voting) but differed between architectures ($\rho = 0.18$), while total attribution to the collinear blood-pressure pair was nearly identical. All results are a computational benchmark on labels without clinical adjudication and are not evidence of diagnostic validity. Code, split identifiers and observation-level predictions are provided.

Keywords: sleep disorder; insomnia; sleep apnea; stacking ensemble; XGBoost; duplicate leakage; calibration; explainable machine learning

1 Introduction

Structured sleep-health tables are inexpensive to analyse and are widely used to benchmark machine-learning methods, but their simplicity can conceal weaknesses in outcome definition, data provenance, duplicated records and evaluation design. Strong accuracy on a few hundred rows is more often a statement about the dataset than about the method. Stacked generalisation [30] is a standard way to combine heterogeneous learners, and its validity depends on how the second-level training data are built [10, 18].

Our earlier draft reported 97.33% accuracy for an RF–XGB stack on a single 75-observation hold-out and a one-observation advantage over soft voting. Reviewers rightly asked whether the 219 "no disorder"

labels were genuine or missing, whether duplicates and tuning had inflated the result, whether the supplementary analyses evaluated the same model as the main text, and whether any of it could be reproduced. This revision answers those questions with re-analysis rather than argument. We make five contributions.

1. **Data audit.** We verify outcome coding against the source file, quantify exact duplicates and label conflicts, and derive the accuracy ceiling they impose (Section 3).
2. **Leakage-aware, consistent evaluation.** The primary RF–XGB stack, soft voting, their base learners, simple baselines and six further learners are evaluated under identical repeated, duplicate-aware splits, with tuning confined to training partitions (Section 4.2).
3. **Quantified protocol effects.** We measure how much plain stratified splitting and a single hold-out overstate performance on this file (Sections 5.1 and 5.2).
4. **Paired inference and calibration.** Differences are tested with cluster bootstrap, a corrected resampled t-test and McNemar tests, and probabilities are assessed with Brier score, log-loss and reliability plots (Sections 5.3–5.5).
5. **Explanation audit.** SHAP stability, the effect of predictor collinearity across architectures, and surrogate fidelity of LIME are measured rather than assumed (Section 5.7).

We do not claim a new learning algorithm. The stack is the standard RF/XGBoost/logistic-regression construction with encoded-predictor pass-through; the contribution is a verified implementation of it and an honest account of what it does and does not achieve on this benchmark. We also revise the framing: the task is classification of a *recorded* status, not risk prediction or clinical decision support.

2 Related work

Machine learning for sleep disorders concentrates on apnea screening from electrocardiography, oximetry, respiratory and wearable signals [1, 3, 12, 29]; these studies differ from the low-dimensional lifestyle table used here, which has neither waveforms nor polysomnographic indices, and consistently separate automated screening from clinical diagnosis. For tabular data, tree ensembles remain highly competitive with neural methods, particularly on small samples [4, 13]. Alom et al. applied ensemble neural networks to the same table [2]; their accuracies come from a different split and are not comparable to ours, so we do not tabulate them. Guidance on reporting and evaluating prediction models emphasises provenance, calibration, uncertainty, internal validation that mirrors deployment, and external validation [7, 8, 21, 27]; we follow it where the data permit.

3 Data and audit

3.1 Source, provenance and outcome coding

The data are the public Kaggle Sleep Health and Lifestyle Dataset [17]: 374 rows, a Person ID, 11 predictors (gender, age, occupation, sleep duration, quality of sleep, physical activity level, stress level, BMI category, blood pressure, heart rate, daily steps) and the outcome *Sleep Disorder*. The source gives no collection protocol, and the structure we report below (templated, repeated records) is consistent with synthetic or heavily templated generation. We therefore treat it as a methodological benchmark, not as an observational cohort, and make no prevalence or clinical claims.

Outcome coding. Opening the raw CSV shows that the Sleep Disorder column contains exactly three literal strings: `None` (219), `Sleep Apnea` (78) and `Insomnia` (77); there are no empty fields. The earlier

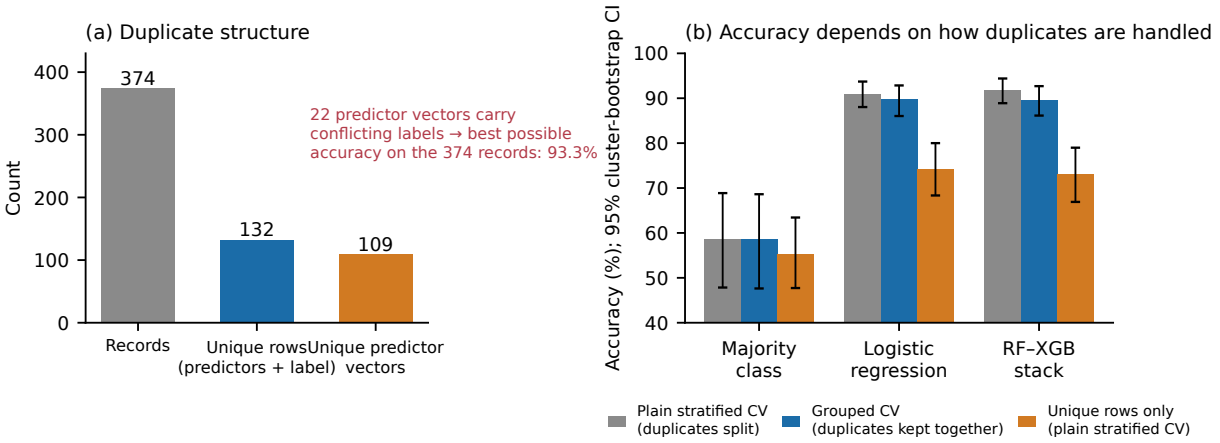


Figure 1: (a) Record, unique-row and unique-predictor-vector counts. (b) Accuracy of the majority class, logistic regression and the RF–XGB stack under three protocols (10 repeated 5-fold CV; error bars are 95% cluster-bootstrap CIs, see Section 4.3).

description of 219 "blank or missing" outcomes arose because pandas, by default, parses the string `None` as a missing value. All analyses here read the file with `keep_default_na=False`, and the code asserts that no missing values remain and that the three labels are present. The 219 records are therefore a genuine "None" category as defined by the dataset, not missing outcomes; *what* "None" means clinically (no disorder, no diagnosis recorded, or not assessed) is undocumented and is discussed as a limitation.

3.2 Duplicates, label conflicts and the accuracy ceiling

After removing Person ID, 242 of the 374 rows are exact duplicates of an earlier row (including the label), leaving 132 unique rows (None 73, Insomnia 29, Sleep Apnea 30). Ignoring the label, there are only 109 distinct predictor vectors, and 22 of them (covering 128 records) appear with more than one label. No deterministic function of the predictors can therefore exceed 349/374 = 93.3% accuracy on this file. Fig. 1a summarises this structure.

3.3 Predictors

Occupation was retained as a contextual predictor but may encode socioeconomic structure and transport poorly. The synonymous BMI labels "Normal" and "Normal Weight" were merged; blood pressure was split into systolic and diastolic values. Sleep duration and quality overlap conceptually with the definition of insomnia, so the predictor–outcome relation is partly circular; Section 5.6 measures how much performance depends on them.

4 Methods

4.1 Models

Primary model (RF–XGB stack). Algorithm 1 defines the primary two-learner stack. Preprocessing (standardisation of nine numeric predictors; one-hot encoding of gender, occupation and BMI category with the vocabulary taken from the training partition) is refitted inside every inner fold. The meta-learner is a multinomial logistic regression trained on $[x', p_{\mathrm{RF}}, p_{\mathrm{XGB}}]$, where x' is the encoded predictor vector passed through, and $p_{\mathrm{RF}}, p_{\mathrm{XGB}}$ are out-of-fold class probabilities. Inner folds are stratified and *group-aware*, so identical records never straddle an inner split. The base learners are refitted on the full training partition for

Algorithm 1 Fold-isolated RF–XGB stacking (training partition D_{tr}, new record x)

Require: raw training data (X_{tr}, y_{tr}); $K = 5$ stratified, group-aware folds

Initialise out-of-fold matrices $P_{\mathrm{RF}}, P_{\mathrm{XGB}} \in \mathbb{R}^{n_{tr} \times 3}$ and pass-through matrix T
for $k = 1, \ldots, K$ **do**
    fit scaler and encoder τ_{-k} on the folds other than k only
    fit RF and XGBoost on $\tau_{-k}(X_{-k})$; store their probabilities for fold k in $P_{\mathrm{RF}}, P_{\mathrm{XGB}}$
    store $\tau_{-k}(X_k)$ in T at the same row indices
end for
fit multinomial logistic regression g on $Z = [T, P_{\mathrm{RF}}, P_{\mathrm{XGB}}]$ with labels y_{tr}
fit τ, RF and XGBoost on all of D_{tr}
return $p(x) = g\big([\tau(x), f_{\mathrm{RF}}(\tau(x)), f_{\mathrm{XGB}}(\tau(x))]\big)$, $\hat{y} = \arg\max_c p_c(x)$

inference. **Fixed configuration:** RF with 100 trees, depth 10, minimum split 5; XGBoost with 800 rounds, learning rate 0.02, depth 3, subsample/column sample 0.9, minimum child weight 3, $\gamma = 0.15$, $\lambda = 1.2$; meta-learner $C = 2.0$. These values are taken from the earlier draft; their provenance cannot be verified, so Section 5.4 repeats the evaluation with all tuning inside the training partitions.

Comparators. All comparators receive the same encoded training data: majority class; logistic regression ($C = 1$); decision tree (depth 3); RF and XGBoost as configured above; soft voting, the unweighted mean of the *same fitted* RF and XGBoost probabilities; and, as further competitive learners, extra trees, scikit-learn HistGradientBoosting, LightGBM [19], CatBoost [25], an RBF SVM, and a two-layer MLP. The SVM appears only as an independent baseline; no analysis uses a three-learner stack. Hyper-parameters of these additional learners are modest fixed defaults (Appendix A) and are not tuned, which disadvantages them on very small samples (Section 5.2).

4.2 Evaluation protocols

All protocols use 5-fold cross-validation repeated 10 times with different seeds, and every model is scored on identical outer splits (workflow in Appendix B).

- **Grouped CV (primary).** Records with identical predictor vectors form a group; folds are stratified by class and grouped, so no predictor vector appears in both training and test folds (374 records).
- **Plain stratified CV.** The same procedure without grouping, which lets duplicates fall on both sides of a split (the leakage-prone alternative).
- **Unique rows.** The 132 unique rows only, with plain stratified splitting (a sensitivity analysis free of exact duplicates; 22 conflicting predictor vectors remain).
- **Nested tuning.** For grouped CV and unique rows (5 repeats each), an inner 3-fold grouped CV on the outer-training partition selects the main hyper-parameter(s) of each model by log-loss (logistic $C \in \{0.1, 1, 10\}$; RF depth $\in \{3, 6, 10\}$; XGBoost rounds $\in \{200, 800\}$; soft voting: RF depth and XGBoost rounds jointly; stack: XGBoost rounds and meta-learner $C \in \{0.1, 1, 10\}$, with RF depth fixed at 10 for cost). The outer test fold is never used for selection. For the unique-rows protocol the nested outer and inner splits are grouped by predictor vector, whereas the non-nested unique-rows analysis uses plain stratified splits, so Tables 3 and 5 are not strictly like-for-like.
- **Predictor ablation.** Grouped 10×5-fold CV with the same fixed models after removing sleep duration and quality, and additionally stress level.
- **Original split.** A re-implementation of the single stratified 80:20 split (seed 42; 299/75), including observation-level output.

4.3 Metrics and statistical comparison

We report accuracy, macro-F1, macro one-vs-rest AUC [11], multiclass Brier score [5], log-loss and a five-bin equal-mass top-label expected calibration error (ECE_5; unstable at this sample size, so interpreted with caution) [14]. Repeat-level results are averaged over the 10 repeats. Confidence intervals for accuracy are *cluster bootstrap* intervals (5,000 resamples of predictor-vector groups, applied to repeat-averaged correctness), which reflect sampling uncertainty of the data; the spread across repeats alone would understate it. For stack versus each comparator we report the mean accuracy difference with a 20,000-resample cluster-bootstrap interval, the Nadeau–Bengio corrected resampled t-test over the 50 fold-level differences [23], and an exact McNemar test [9] in each repeat (median p and the number of repeats with $p < 0.05$). Single-split intervals are exact Clopper–Pearson intervals [6]. We do not adjust for multiplicity; with 12 comparators, isolated nominal findings should be read as exploratory.

4.4 Explanation analyses

SHAP values were obtained with the permutation explainer [22], using a 25-record background sampled from the training partition and 35 explanation records sampled from the test partition of the original split, with $2p+1$ evaluations ($p = 25$ encoded features), on both the stack and the soft-voting model. The procedure was repeated 10 times with independent backgrounds and explanation samples; one-hot columns were summed per original predictor before taking absolute values, and importance is the mean of $|\phi|$ over records and classes. Stability is the mean pairwise Spearman correlation of the 12 importance values across runs and the Jaccard overlap of the top-3 sets. To examine collinearity we report Spearman correlations and variance-inflation factors on all 374 records, and a held-out permutation test (grouped 5-fold, 10 permutations) that shuffles a single predictor or a correlated block jointly and records the increase in log-loss. LIME [26] was run in the encoded space (5,000 perturbations, 10 seeds) for four held-out records; for each we report the identifier, reference label, predicted probabilities, the weighted-surrogate R^2 (fidelity) and the rank agreement of weights across seeds.

4.5 Software and reproducibility

Python 3.13, scikit-learn 1.9.1 [24], XGBoost 3.4.1, LightGBM 4.7.0, CatBoost 1.2.10, SHAP 0.54.1, LIME 0.2.0.1, pandas 3.0.5, NumPy 2.5.3, SciPy 1.18.1. Seeds are fixed per repeat. The supplementary archive contains the complete code, a script that regenerates every table and in-text number from the stored results, the split identifiers (repeat, fold, row) for all protocols, and observation-level labels, predicted classes and class probabilities for every model, repeat and fold (Data and code availability).

5 Results

5.1 Reproducing the original single hold-out

Re-running the documented specification on the original 80:20 split (seed 42) gave 72/75 correct for the stack (96.00%, exact 95% CI 88.8–99.2%; macro-F1 0.94) and 72/75 for soft voting (Table 1); using plain stratified rather than group-aware inner folds also gave 72/75. We were therefore *unable to reproduce the 73/75 (97.33%) reported in the earlier draft*, and we withdraw that figure; the original analysis code was not available for inspection. On this split the stack and soft voting differ on one record each way (McNemar $p = 1.00$; paired bootstrap CI for the difference -4.0 to $+4.0$ pp). Training accuracy is 91.6%, close to test accuracy, which is expected rather than reassuring: 68 of the 75 test records have a predictor vector that also occurs in the training set, so the test set largely re-asks questions the models have already seen. On

Table 1: Re-implementation of the single stratified 80:20 hold-out (seed 42). The last column counts test records whose predictor vector never occurs in the training set.

Model	Correct/75	Accuracy (%) [exact 95% CI]	Macro-F1	Train acc. (%)	Unseen-vector rows correct
Logistic regression	71	94.7 [86.9, 98.5]	0.92	91.0	6/7
Random forest	72	96.0 [88.8, 99.2]	0.94	92.0	6/7
XGBoost	70	93.3 [85.1, 97.8]	0.89	91.6	4/7
Soft voting (RF+XGB)	72	96.0 [88.8, 99.2]	0.94	91.6	6/7
Stacking (RF+XGB)	72	96.0 [88.8, 99.2]	0.94	91.6	5/7

Table 2: Grouped repeated CV (10×5-fold; 374 records; identical splits for all models; fixed hyper-parameters). Accuracy CIs are cluster-bootstrap; the other columns are means over 10 repeats.

Model	Accuracy (%) [95% CI]	Macro-F1	Macro AUC	Brier	Log-loss	ECE_5
Majority class	58.6 [47.6, 68.6]	0.25	0.50	0.571	0.966	0.006
Decision tree (depth 3)	88.1 [84.1, 91.4]	0.85	0.90	0.209	0.775	0.044
Logistic regression	89.7 [86.0, 92.8]	0.87	0.92	0.188	0.435	0.063
SVM (RBF)	90.0 [86.7, 92.9]	0.87	0.93	0.172	0.369	0.027
MLP (32,16)	89.7 [86.5, 92.6]	0.87	0.91	0.188	0.509	0.061
Random forest	89.8 [86.4, 92.9]	0.87	0.92	0.191	1.119	0.079
Extra trees	88.7 [84.8, 92.0]	0.86	0.92	0.207	1.337	0.075
HistGradientBoosting	89.0 [85.6, 92.1]	0.85	0.91	0.207	0.678	0.080
LightGBM	89.5 [86.1, 92.5]	0.86	0.91	0.204	0.648	0.088
CatBoost	89.8 [86.3, 92.7]	0.87	0.92	0.180	0.397	0.045
XGBoost	89.8 [86.3, 92.9]	0.87	0.92	0.185	0.431	0.055
Soft voting (RF+XGB)	89.8 [86.3, 92.9]	0.87	0.92	0.184	0.447	0.060
Stacking (RF+XGB)	89.6 [86.1, 92.7]	0.87	0.91	0.186	0.446	0.062

the only 7 test records with unseen vectors the stack was correct on 5, soft voting on 6, logistic regression on 6 (too few to compare). Fitted to all 374 records, RF and the stack reach 93.0% and 93.0% in-sample accuracy against the 93.3% ceiling, which also explains the near-equal training and test accuracies (0.926 and 0.924) in the earlier repeated-split figure.

5.2 Repeated duplicate-aware cross-validation

Table 2 and Fig. 2 give the primary results. The stack reached 89.6% accuracy (95% CI 86.1–92.7; fold-level SD 3.8 pp) and macro-F1 0.87. Every model with real capacity landed within about one point: soft voting 89.8%, random forest 89.8%, XGBoost 89.8%, logistic regression 89.7%, CatBoost 89.8%, SVM 90.0%; a depth-3 tree reached 88.1% and the majority class 58.6%. A shallow tree and a linear model recover almost everything because the table is dominated by a few templated patterns (occupation, BMI, blood pressure, age).

Protocol effects. Plain stratified CV, which lets duplicates straddle splits, raised the stack's accuracy to 91.8% (2.2 points above grouped CV). On the 132 unique rows the stack reached 73.0% (CI in Table 3), 16.6 points below grouped CV, against 55.3% for the majority class; logistic regression reached 74.1%. Several untuned learners (extra trees, LightGBM, the MLP) fell to near majority-class accuracy on 132 rows, a reminder that default settings do not transfer to very small samples. Performance on this table is thus largely a function of how duplicates are treated.

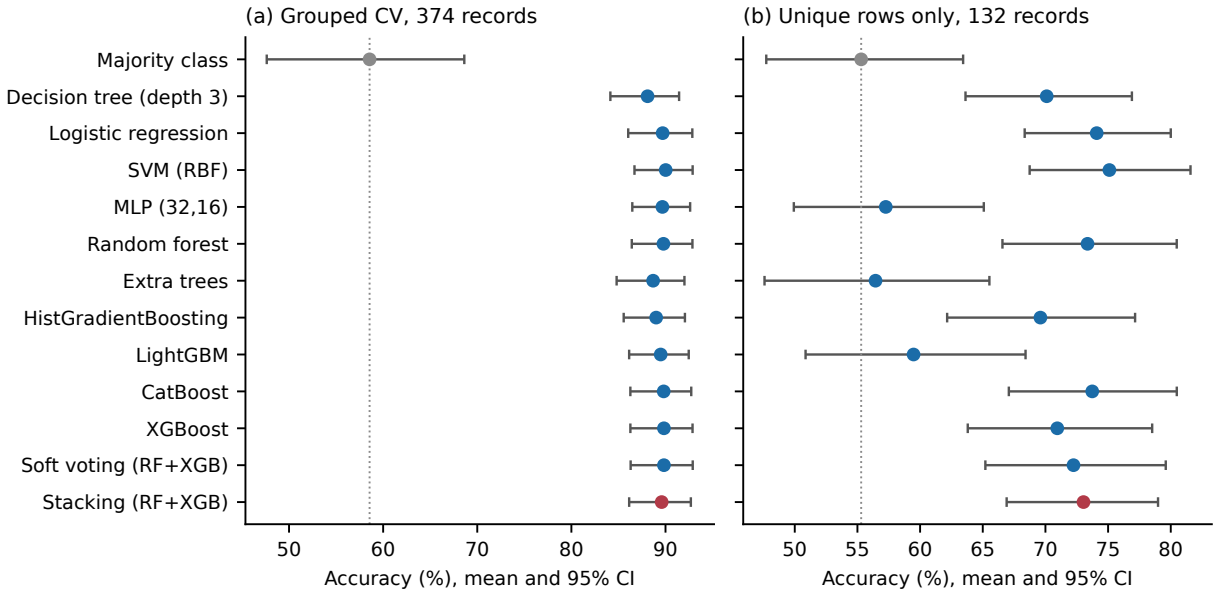


Figure 2: Accuracy of all models under grouped CV (a) and on the unique rows (b). Points are 10-repeat means; bars are 95% cluster-bootstrap CIs; the dotted line is the majority-class accuracy. The stack is shown in red.

5.3 Paired comparisons

Table 4 and Fig. 3 test the stack against every comparator. Against soft voting under grouped CV the mean difference is -0.2 pp (95% CI -1.8 to +1.1; median McNemar $p = 0.68$; stack better, tied and worse in 15/17/18 of 50 folds; on average 4.2 records were correct only under the stack and 5.1 only under soft voting). Differences against logistic regression (-0.1 pp, CI -1.2 to +0.9), RF (-0.2) and XGBoost (-0.2) are likewise indistinguishable from zero; apart from the majority-class baseline, the only interval that excludes zero is that against the Decision tree (depth 3), a deliberately weak baseline. On unique rows the intervals are wider and include zero for all competitive learners (extra trees, LightGBM and the MLP are worse than the stack, but they are untuned and fall to near majority-class accuracy; against soft voting: +0.8 pp, CI -2.8 to +4.5). *The stack is not superior to soft voting, to its own base learners, or to logistic regression on this dataset.*

5.4 Nested tuning

When all hyper-parameter selection is confined to inner folds of the training partition, the picture is unchanged (Table 5). Under grouped CV the stack reaches 90.0% (CI 86.6–93.1), soft voting 89.9%, XGBoost 90.1%, RF 89.2% and logistic regression 89.0%. The stack minus soft voting difference is +0.2 pp (CI -1.0 to +1.4); minus XGBoost -0.1 (-1.7 to +1.5); minus logistic regression +1.0 (-0.2 to +2.5); minus RF +0.8 (+0.1 to +1.8). The RF interval narrowly excludes zero, an isolated nominal result among eight comparisons that we do not interpret as evidence of an advantage. On unique rows the stack reaches 77.2% (71.5–83.2) with every paired interval including zero except logistic regression (+2.3 pp; +0.7 to +4.2), and no repeat-level McNemar test was significant in any comparison. Selection sat at the lower edge of the grid in nearly every fold (RF depth 3 and XGBoost with 200 rounds in all 25 outer folds in both protocols; logistic $C = 0.1$ in most), so tuning acted mainly as extra regularisation, and a wider grid might select differently.

Table 3: Repeated CV on the 132 unique rows only (plain stratified 5-fold, 10 repeats). Same format as Table 2.

Model	Accuracy (%) [95% CI]	Macro-F1	Brier	Log-loss
Majority class	55.3 [47.7, 63.4]	0.24	0.594	0.998
Decision tree (depth 3)	70.1 [63.6, 76.9]	0.64	0.488	2.904
Logistic regression	74.1 [68.3, 80.0]	0.68	0.435	0.838
SVM (RBF)	75.1 [68.7, 81.6]	0.68	0.405	0.740
MLP (32,16)	57.3 [49.9, 65.1]	0.52	0.723	3.316
Random forest	73.4 [66.6, 80.5]	0.66	0.482	1.885
Extra trees	56.4 [47.6, 65.5]	0.48	0.808	9.056
HistGradientBoosting	69.6 [62.2, 77.2]	0.63	0.515	1.126
LightGBM	59.5 [50.9, 68.4]	0.52	0.726	1.837
CatBoost	73.7 [67.1, 80.5]	0.67	0.460	0.862
XGBoost	70.9 [63.8, 78.5]	0.64	0.486	0.941
Soft voting (RF+XGB)	72.2 [65.2, 79.6]	0.65	0.478	0.977
Stacking (RF+XGB)	73.0 [66.9, 79.0]	0.67	0.452	0.900

5.5 Probability calibration

Fig. 4 shows reliability and Brier scores for grouped CV. Brier scores (lower is better) are similar for the stack (0.186), soft voting (0.184), XGBoost (0.185) and logistic regression (0.188); the SVM and CatBoost were best (0.172, 0.180). Log-loss is similar for XGBoost (0.431), logistic regression (0.435), the stack (0.446) and soft voting (0.447), and best for the SVM (0.369) and CatBoost (0.397). Random forest stands out (1.119) because it assigns exactly zero probability to the true class for about 3% of held-out predictions, which log-loss penalises heavily; averaging with XGBoost removes this in soft voting and stacking. ECE_5 was 0.062 for the stack and 0.060 for soft voting. No post-hoc recalibration was applied, and the five-bin estimate is noisy.

5.6 Dependence on sleep-behaviour predictors

Removing sleep duration and quality, or additionally stress, barely changes accuracy (Table 6). The signal is carried by other predictors (occupation, BMI, blood pressure, age), which is consistent with templated generation. The conceptual overlap between sleep duration/quality and insomnia is therefore not what drives the benchmark accuracy, although it would matter for any genuine deployment.

5.7 Explanation stability, collinearity and local audit

Stability. Across 10 independent backgrounds and explanation samples, SHAP importance rankings were highly reproducible within an architecture (mean pairwise rank correlation 0.98 for stacking, 0.97 for soft voting; minimum 0.94 and 0.94), although top-3 sets were less stable (Jaccard 0.73 and 0.72; Table 7, Fig. 5). Reproducible within a model does not mean the same across models: the mean importance profiles of stacking and soft voting correlate only weakly ($\rho = 0.18$).

Collinearity. The predictors are strongly dependent (Spearman: systolic–diastolic 0.98; sleep duration–quality 0.89; quality–stress -0.91; activity–steps 0.79; variance-inflation factors 35.3 and 38.0 for systolic and diastolic pressure, 13.3 for quality of sleep). The stack passes encoded predictors to a linear meta-learner, whereas soft voting uses trees only, and the two attribute importance differently: soft voting concentrates it on systolic pressure, whereas the stack spreads it across diastolic pressure, occupation, sleep duration and heart rate. The blood-pressure *pair*, however, receives almost the same total attribution under

Table 4: Stack minus comparator, accuracy difference in percentage points with cluster-bootstrap 95% CI; fold-level wins/ties/losses out of 50; per-repeat exact McNemar tests; Nadeau–Bengio corrected t-test. No multiplicity correction.

Comparator	Δ accuracy (pp) [95% CI]	Folds: stack >/=/<	Median McNemar p	Repeats $p < 0.05$	Corrected t-test p
Grouped CV (374 records)					
Majority class	+31.0 [+20.7, +41.8]	50/0/0	<0.001	10/10	<0.001
Decision tree (depth 3)	+1.5 [+0.4, +3.0]	29/14/7	0.18	1/10	0.73
Logistic regression	-0.1 [-1.2, +0.9]	12/24/14	1.00	0/10	0.98
SVM (RBF)	-0.4 [-2.0, +0.8]	12/16/22	0.59	1/10	0.93
MLP (32,16)	-0.1 [-0.9, +0.7]	13/16/21	0.48	0/10	0.98
Random forest	-0.2 [-1.6, +1.2]	16/16/18	0.90	0/10	0.97
Extra trees	+0.9 [-0.2, +2.2]	23/14/13	0.53	2/10	0.85
HistGradientBoosting	+0.6 [-1.2, +2.3]	21/12/17	0.24	1/10	0.93
LightGBM	+0.1 [-1.4, +1.4]	11/24/15	0.32	0/10	0.99
CatBoost	-0.2 [-1.6, +1.1]	14/19/17	0.68	0/10	0.96
XGBoost	-0.2 [-1.8, +1.2]	14/19/17	0.77	1/10	0.95
Soft voting (RF+XGB)	-0.2 [-1.8, +1.1]	15/17/18	0.68	0/10	0.95
Unique rows only (132 records)					
Majority class	+17.7 [+10.7, +25.2]	48/1/1	<0.001	10/10	0.31
Decision tree (depth 3)	+3.0 [-0.2, +6.3]	30/6/14	0.40	2/10	0.88
Logistic regression	-1.1 [-2.3, +0.1]	9/23/18	1.00	1/10	0.92
SVM (RBF)	-2.0 [-4.8, +0.7]	11/18/21	0.75	2/10	0.85
MLP (32,16)	+15.8 [+12.0, +19.5]	45/4/1	<0.001	10/10	0.51
Random forest	-0.3 [-3.3, +2.6]	17/15/18	0.54	0/10	0.98
Extra trees	+16.7 [+11.8, +21.2]	48/0/2	<0.001	10/10	0.45
HistGradientBoosting	+3.5 [-0.3, +7.4]	30/9/11	0.47	1/10	0.83
LightGBM	+13.6 [+8.5, +18.6]	45/1/4	0.001	9/10	0.53
CatBoost	-0.7 [-3.6, +2.2]	18/10/22	1.00	0/10	0.96
XGBoost	+2.1 [-1.8, +6.1]	28/6/16	0.33	0/10	0.90
Soft voting (RF+XGB)	+0.8 [-2.8, +4.5]	21/13/16	0.42	0/10	0.96

both architectures (0.148 versus 0.156), and permuting the pair jointly on held-out data increases log-loss far more than permuting either member alone (stack: 0.647 versus 0.125 and 0.298; soft voting: 0.318 versus 0.136 and 0.015). In other words, the architectures disagree about *which* of two near-duplicate variables to credit, not about whether blood pressure matters; single-variable rankings of correlated predictors should not be interpreted individually. The sleep/stress block, in contrast, differs between architectures (0.148 versus 0.054), because the linear meta-learner uses these predictors directly whereas the trees barely do.

Local audit. Table 8 identifies each explained record by Person ID, with its reference label, predicted probabilities and surrogate fidelity. LIME R^2 was 0.76–0.79 for the stack and 0.51–0.64 for soft voting: good for the stack and clearly weaker for soft voting, whose local surrogate should therefore be read more cautiously. Rank agreement of weights across seeds was at least 0.97 for the stack. The record with Person ID 84 is a "None" record predicted with probability only 0.38 by the stack, and Person ID 19 is a misclassified insomnia record (predicted apnea with probability 0.67 by both models). Local explanations differ between architectures for the same record, as the global ones do.

6 Discussion

Principal findings. On this benchmark, a leakage-controlled RF–XGB stack achieves roughly 90% accuracy under duplicate-aware validation, but so do a logistic regression, a shallow tree (within about 1.5 points) and every competitive tree or kernel learner. Stacking brings no measurable gain over soft voting, its own base learners or the linear baseline, neither in accuracy nor in calibration. The one-observation difference in the earlier draft was not evidence of an advantage and, as we now show, the headline figure

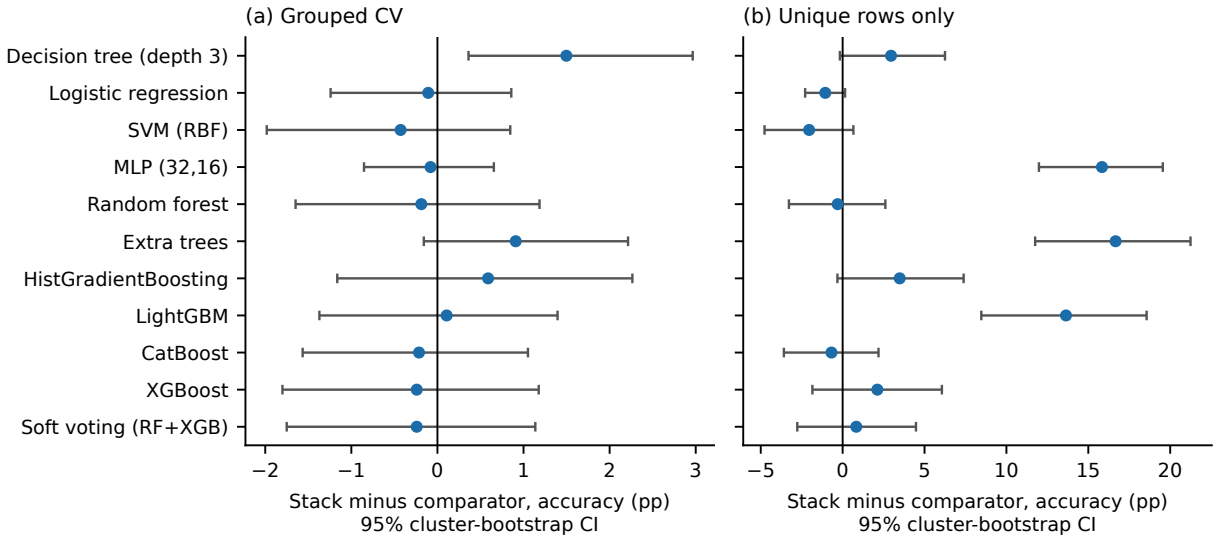


Figure 3: Accuracy difference (stack minus comparator) with 95% cluster-bootstrap CI under grouped CV (a) and on unique rows (b). Positive values favour the stack.

Table 5: Repeated nested CV (5 repeats × 5 outer folds; inner 3-fold grouped selection on training partitions only).

Model	Accuracy (%) [95% CI]	Macro-F1	Brier	Log-loss
Grouped CV (374 records)				
Logistic regression	89.0 [85.3, 92.4]	0.86	0.198	0.428
Random forest	89.2 [85.5, 92.5]	0.86	0.181	0.385
XGBoost	90.1 [86.7, 93.2]	0.87	0.175	0.378
Soft voting (RF+XGB)	89.9 [86.5, 93.0]	0.87	0.175	0.374
Stacking (RF+XGB)	90.0 [86.6, 93.1]	0.87	0.184	0.399
Unique rows only (132 records)				
Logistic regression	75.0 [69.0, 81.0]	0.70	0.415	0.755
Random forest	76.7 [70.6, 82.9]	0.71	0.394	0.730
XGBoost	76.9 [70.7, 83.2]	0.71	0.394	0.737
Soft voting (RF+XGB)	76.5 [70.4, 82.5]	0.71	0.390	0.725
Stacking (RF+XGB)	77.2 [71.5, 83.2]	0.72	0.406	0.741

itself was not reproducible.

Why the earlier estimate was optimistic. Three mechanisms compound: duplicated rows allow memorisation across a random split (68 of 75 test records here); a single 75-record split has an exact 95% interval of roughly 88.8–99.2%; and fixed hyper-parameters of unverifiable origin were used. Accuracy on unique rows (73.0% for the stack) is a fairer indication of how the models behave on records they have not effectively seen, and even that remains conditional on a dataset with conflicting labels. Because the file caps attainable accuracy at 93.3%, near-ceiling accuracy on it indicates memorisation of its generating patterns, not clinical signal.

Statistical power. With 75 test records, the single-split comparison between stack and soft voting rests on two discordant records; an exact McNemar test cannot reject with so few discordant pairs, and at the observed discordance rate the smallest difference detectable with 80% power is about 5.3 points, four times the one-record (1.33-point) difference reported in the earlier draft. Detecting a true 1.33-point difference at that discordance rate would require roughly 1180 test records, more than three times the entire dataset. Even the repeated, paired analysis here uses 374 correlated records and 109 distinct predictor vectors, so its intervals (about ±1.5 pp for most comparisons) cannot exclude advantages or disadvantages of that size.

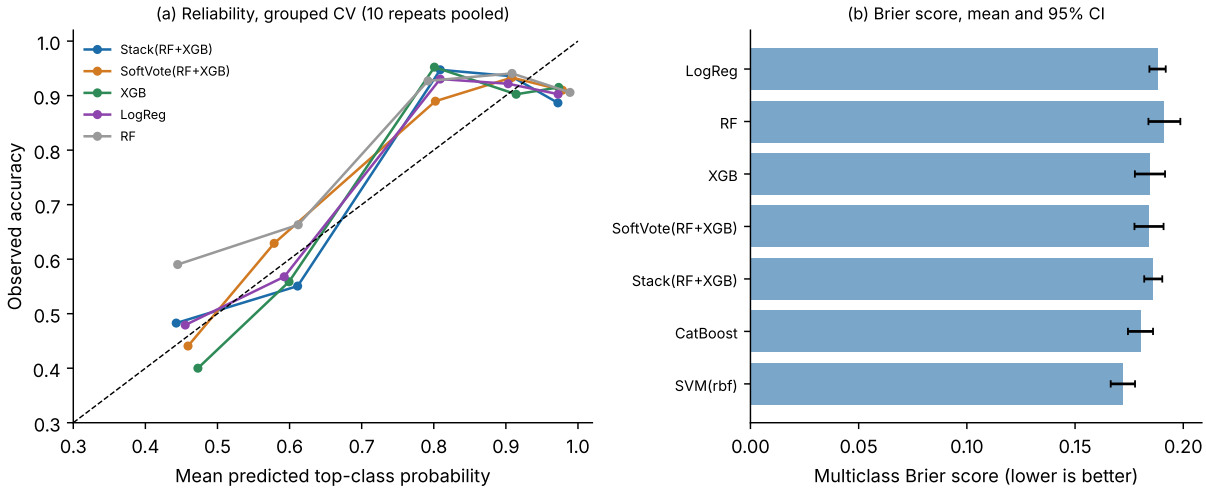


Figure 4: (a) Reliability of top-class probabilities (bins with at least 20 predictions; 10 grouped-CV repeats pooled). (b) Multiclass Brier score, mean and 95% t-interval across repeats.

Table 6: Grouped repeated CV accuracy (%, 10 repeats) after removing sleep-behaviour predictors.

Predictor set	Stacking	Soft voting	Logistic regression
All predictors	89.6	89.8	89.7
Without sleep duration & quality	89.9	89.9	89.1
Without sleep duration, quality & stress	89.2	89.9	88.9

The correct reading is "no evidence of a difference", not "evidence of no difference".

Collinearity and explanations. Section 5.7 shows that attribution rankings of correlated predictors are architecture-specific while block-level attribution is stable. Single-variable SHAP or LIME rankings should therefore be reported together with the correlation structure and, where possible, block-level or grouped permutation importance. Neither technique establishes biological or clinical mechanism: SHAP, LIME and permutation importance describe the behaviour of a fitted function on this data distribution [16, 20].

Benchmark outcomes versus clinical screening. A reader should not treat any figure in this paper as diagnostic evidence. The labels come from an undocumented source without clinical adjudication; the task is reproducing a recorded category from lifestyle and cardiometabolic variables, which is not screening. Real-world screening would require prospectively collected participants with polysomnography or clinician-confirmed outcomes, pre-specified intended use and thresholds, evaluation of sensitivity and specificity at clinically meaningful operating points, calibration in the target population, subgroup performance, and external validation [7, 21, 27]. A model that reproduces a benchmark's labels, even perfectly, may carry no clinical information if those labels were generated by simple rules.

Contribution and novelty. Stacking RF and XGBoost with a logistic meta-learner is standard practice, as are SHAP and LIME; we claim no new algorithm. What this work offers is a verified, leakage-controlled implementation, a quantified account of how duplicate structure and single splits inflate estimates on a widely used benchmark, a like-for-like comparison against simple and recent learners with paired statistics and calibration, and an explanation audit including collinearity and stability. Its practical message is that simple baselines and duplicate checks should precede any claim of ensemble benefit on small tabular health datasets [28].

Table 7: SHAP stability and collinearity diagnostics for the stack and the soft-voting model (original-split training partition; explanation sample 35 test records; 10 runs).

Quantity	Stacking	Soft voting
Mean pairwise rank correlation across 10 runs (min)	0.98 (0.94)	0.97 (0.94)
Mean pairwise Jaccard overlap of top-3 sets	0.73	0.72
Top-3 predictors (mean $\|\phi\|$)	Diastolic BP, Occupation, Sleep Duration	Systolic BP, BMI Category, Physical Activity Level
Blood-pressure block, mean $\|\phi\|$ (SBP+DBP)	0.148	0.156
Sleep/stress block (duration+quality+stress)	0.148	0.054
Activity block (activity+steps)	0.080	0.058
Held-out Δlog-loss, SBP / DBP alone	0.125 / 0.298	0.136 / 0.015
Held-out Δlog-loss, BP block permuted jointly	0.647	0.318

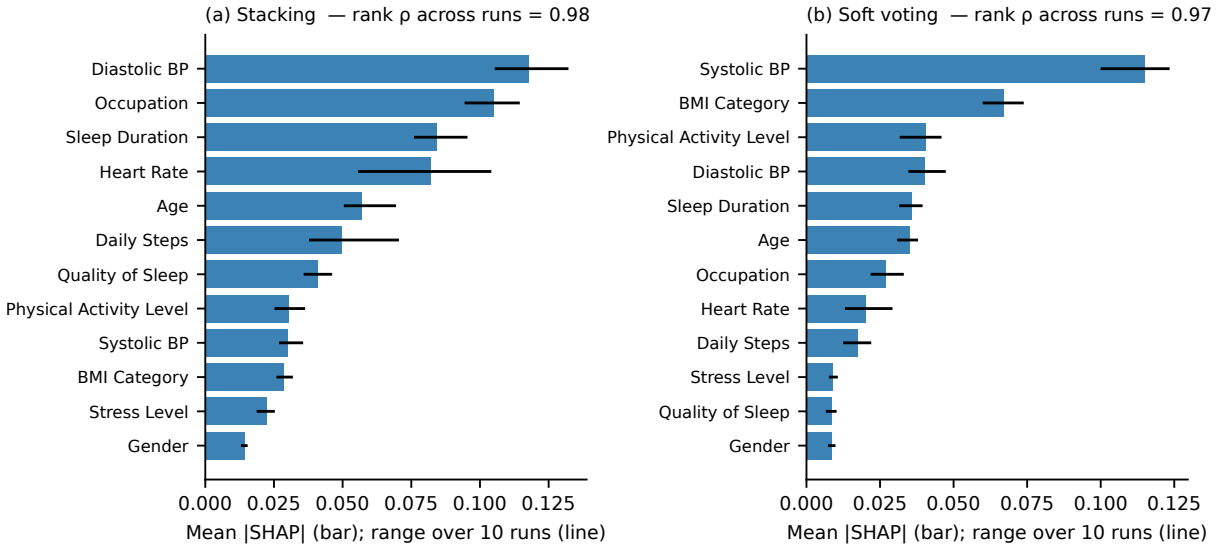


Figure 5: Global SHAP importance (mean $|\phi|$ summed over one-hot columns) for the stack (a) and soft voting (b). Bars: mean over 10 runs; lines: range over runs. Orderings differ between panels.

7 Limitations

(i) The data have no documented provenance and appear templated; results do not generalise to patient populations. (ii) The meaning of the "None" label is undocumented. If "None" means "no disorder recorded" rather than "clinically excluded", the 219 records may contain undiagnosed cases; any classifier then learns to predict the *recording* of a disorder, and class-conditional errors (especially false "None" predictions) are biased towards under-detection in ways this dataset cannot quantify. Treating these records as missing instead would discard 59% of the sample and change the task. (iii) Duplicates and label conflicts impose a ceiling and make group-aware evaluation necessary; the unique-rows analysis retains conflicting vectors. (iv) The sample is small, so even the repeated analyses have wide intervals and repeated splits are not independent; the Nadeau–Bengio correction is approximate and no multiplicity adjustment is applied. (v) The additional learners use fixed, untuned settings. Pretrained tabular foundation models and attention-based tabular networks [15] were not evaluated because pretrained weights were unavailable in our environment, and they would be unlikely to be informative on 109 distinct vectors. (vi) Explanations describe fitted models only. (vii) ECE is unstable at this sample size. (viii) No external cohort was available.

8 Future work

We recommend (a) external validation on a clinically characterised cohort with adjudicated outcomes and polysomnography-based reference standards, with calibration assessed in that population; (b) repeated

Table 8: Audited local explanations (original-split test records). Person ID is the dataset identifier; R^2 is LIME's weighted-surrogate fidelity (mean of 10 seeds); rank ρ is the mean pairwise Spearman correlation of absolute weights across seeds; top-3 features by mean weight in the encoded space.

Model	Person ID	Reference	Predicted	P(Ins./None/Apnea)	LIME R^2	Rank ρ	Top-3 local features
Stacking	244	Insomnia	Insomnia	0.90/0.08/0.02	0.77	0.98	Sleep Duration, Age, Diastolic BP
Soft voting	244	Insomnia	Insomnia	0.97/0.03/0.00	0.51	0.82	Systolic BP, Physical Activity Level, Sleep Duration
Stacking	84	None	None	0.32/0.38/0.30	0.79	0.98	Diastolic BP, Heart Rate, Age
Soft voting	84	None	None	0.07/0.84/0.08	0.64	0.84	Systolic BP, Diastolic BP, BMI Category Normal
Stacking	291	Sleep Apnea	Sleep Apnea	0.06/0.05/0.89	0.76	0.97	Heart Rate, Sleep Duration, Diastolic BP
Soft voting	291	Sleep Apnea	Sleep Apnea	0.02/0.08/0.91	0.60	0.85	Systolic BP, Age, Diastolic BP
Stacking	19	Insomnia	Sleep Apnea	0.08/0.25/0.67	0.76	0.97	Heart Rate, Sleep Duration, Diastolic BP
Soft voting	19	Insomnia	Sleep Apnea	0.28/0.05/0.67	0.60	0.81	Systolic BP, Age, Diastolic BP

nested cross-validation, calibration assessment and paired testing as the default for small tabular studies (applied here); (c) pre-registered sample-size justification for comparisons between ensembles; and (d) a minimum reporting standard for local explanations: the observation identifier, reference label, predicted class probabilities, explainer settings (background size, number of perturbations, kernel width, seeds), surrogate fidelity and stability across seeds, as in Table 8. Evaluating tabular foundation models and tuned neural networks once pretrained weights are available is a natural extension.

9 Conclusion

After verifying outcome coding, removing duplicate leakage from the evaluation and comparing models under identical repeated splits, RF–XGB stacking performs at about the same level as soft voting, its base learners and logistic regression on the Sleep Health and Lifestyle Dataset (89.6% accuracy under grouped CV; 73.0% on unique rows). The earlier single-split result (97.33%) was not reproducible and overstated performance. SHAP rankings are reproducible within a model but architecture-dependent for correlated predictors. These findings concern a computational benchmark; they say nothing about diagnostic validity.

Declarations

Data and code availability. The dataset is publicly available on Kaggle [17]; reuse is subject to the platform's licence and terms, so the file itself is not redistributed (its SHA-256 and row count are checked by the code). The supplementary archive contains the code, software versions, seeds, split identifiers for all protocols, observation-level reference labels, predicted classes and class probabilities for every model, repeat and fold, the SHAP and LIME run outputs, and the scripts that regenerate every table and in-text number.

Ethics. Secondary computational analysis of a public dataset; no participants were recruited or contacted and no new human or animal data were collected. Person ID is used only to cite audited records.

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Declaration of competing interest. The authors declare no known competing financial interests or personal relationships that could have influenced the work reported here.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process. During the preparation of this work the authors used AI assistants (Claude, Anthropic) to assist with code development for the revision analyses, manuscript editing and LaTeX formatting. The authors reviewed and edited the output and take full responsibility for the content of the publication.

Author contributions. [To be completed by the authors.]

A Hyper-parameters of all models

Table 9: Fixed configurations (all other settings are library defaults). Seeds: 1000+repeat index for repeated CV and the predictor ablation, 2000+repeat index for nested CV, the repeat index for outer splits, 42 for the original split, 7 (with per-run seeds) for SHAP and 0–9 for LIME; see code.

Model	Configuration
Logistic regression	multinomial, L2, $C = 1$, lbfgs, max 5000 iterations
Decision tree	depth 3
Random forest	100 trees, depth 10, min. split 5, min. leaf 1
XGBoost	800 rounds, learning rate 0.02, depth 3, subsample 0.9, colsample 0.9, min. child weight 3, $\gamma = 0.15$, $\lambda = 1.2$, hist
Soft voting	mean of the fitted RF and XGBoost probabilities
Stack	RF and XGBoost as above; 5 stratified group-aware inner folds; meta-learner multinomial logistic, $C = 2.0$, inputs [encoded x, p_{RF}, p_{XGB}]
Extra trees	300 trees
HistGradientBoosting	200 iterations, learning rate 0.05
LightGBM	300 rounds, learning rate 0.05, 8 leaves, min. child samples 5
CatBoost	300 iterations, depth 4, learning rate 0.05
MLP	hidden layers (32, 16), $\alpha = 10^{-3}$, max 3000 iterations
SVM	RBF, $C = 2$, Platt probabilities

B Workflow

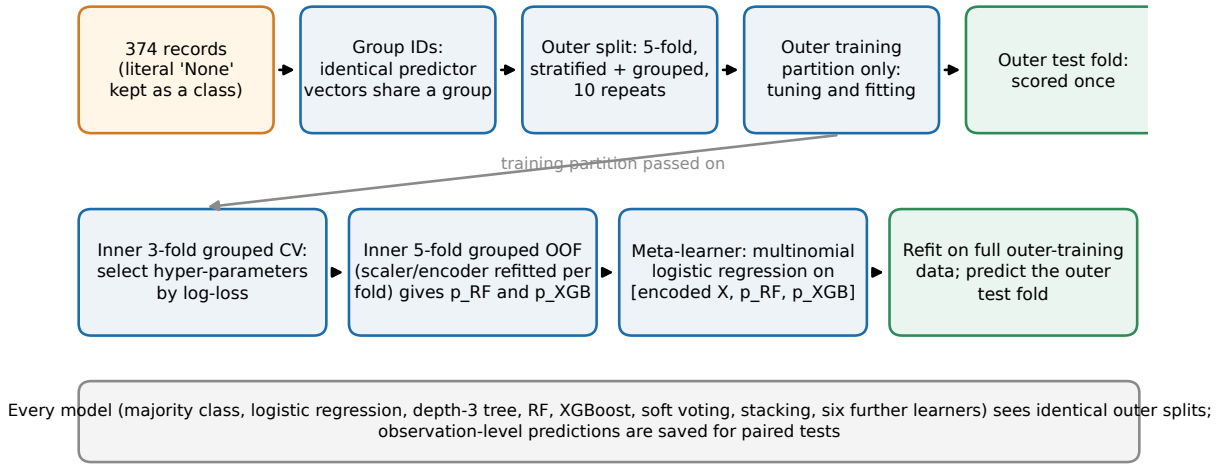


Figure 6: Evaluation workflow for the repeated nested protocol. Grouping, outer splitting, inner tuning, fold-isolated preprocessing and outer-fold scoring are applied identically to every model.

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